GO_Biological_Process_2023 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Intermediate Filament Organization (GO:0	−0.21152670	65	3.774e−09	2.041e−05	KRT26:2 KRT2:3 KRT76:4 KRT34:9 KRT75:20 KRT83:24
Sensory Perception Of Smell (GO:0007608)	−0.11534385	208	1.067e−08	2.884e−05	OR2A14:8 OR10G9:16 OR51V1:62 OR8D4:66 OR2AG1:85 OR51I2:89
Sensory Perception Of Chemical Stimulus	−0.14629973	98	5.750e−07	1.036e−03	CALHM3:21 UGT2A1:59 OR51V1:62 OR8D4:66 OR51I2:89 OR5D3P:230
Regulation Of Transcription By RNA Polym	0.03399052	1948	1.400e−06	1.892e−03	PARP9:8 PHOX2A:22 CHD3:28 ZNF24:31 GADD45A:39 PROP1:45
Mitochondrial Gene Expression (GO:014005	0.12516256	102	1.283e−05	1.388e−02	GADD45GIP1:53 GFM2:288 MRPL54:328 MRPL47:359 MRPL41:384 MRPL2:513
Regulation Of DNA−templated Transcriptio	0.03017590	1857	2.694e−05	2.428e−02	ZNF628:1 WNT1:2 PARP9:8 PHOX2A:22 CHD3:28 ZNF24:31
Mitochondrial Translation (GO:0032543)	0.11923727	97	5.036e−05	3.026e−02	GADD45GIP1:53 GFM2:288 MRPL54:328 MRPL47:359 MRPL41:384 MRPL2:513
Positive Regulation Of DNA−templated Tra	0.03539101	1209	4.579e−05	3.026e−02	WNT1:2 PARP9:8 PHOX2A:22 CHD3:28 ZNF24:31 SOX17:66
Regulation Of Gene Expression (GO:001046	0.03725085	1085	4.430e−05	3.026e−02	ZNF628:1 RFPL4B:7 PARP9:8 CHD3:28 RBM12B:41 PDCD5:42
Negative Regulation Of Gene Expression (	0.06308747	332	8.327e−05	4.093e−02	PARP9:8 ITGB8:47 CSNK2A1:79 POU4F1:150 TIA1:176 CAPRIN1:253
Positive Regulation Of Transcription By	0.03917508	908	7.638e−05	4.093e−02	PHOX2A:22 ZNF24:31 SOX17:66 AKAP8L:106 TBX5:112 TFR2:136
Protein K6−linked Ubiquitination (GO:008	−0.37016964	9	1.203e−04	5.420e−02	UBE2D4:328 RNF4:756 RNF6:772 UBE2T:819 PRKN:1827 UBE2S:3302
Oxidative Phosphorylation (GO:0006119)	0.13955244	59	2.111e−04	8.780e−02	NDUFS2:234 ATP5F1C:316 NDUFA12:475 NDUFB10:502 NDUFB8:797 NDUFB3:906
Atrial Septum Development (GO:0003283)	0.29326003	13	2.512e−04	9.165e−02	NKX2−5:323 NOTCH2:1077 ZFPM1:1092 GATA4:1282 BMPR2:1580 MDM4:1921
Epidermis Development (GO:0008544)	−0.11556268	84	2.542e−04	9.165e−02	KRT2:3 KRT34:9 KRT83:24 GJB5:58 KRT9:116 SPRR2B:199
Outflow Tract Septum Morphogenesis (GO:0	0.20660757	24	4.598e−04	1.554e−01	NKX2−5:323 TGFBR2:827 BMP4:1269 BMPR2:1580 FGF8:1585 ENG:2365
Cardiac Conduction System Development (G	0.20087240	25	5.089e−04	1.619e−01	TBX5:112 NKX2−5:323 SMAD5:955 NOTCH2:1077 HCN4:1096 GATA4:1282
Aorta Morphogenesis (GO:0035909)	0.23836935	17	6.679e−04	1.642e−01	TGFBR2:827 DLL4:830 JAG1:1358 RBPJ:2274 ENG:2365 BMPR1A:2583
Atrial Septum Morphogenesis (GO:0060413)	0.31134335	10	6.514e−04	1.642e−01	NKX2−5:323 NOTCH2:1077 ZFPM1:1092 GATA4:1282 BMPR2:1580 TBX20:2465
Embryonic Hemopoiesis (GO:0035162)	0.27256150	13	6.675e−04	1.642e−01	FLT3LG:117 SH2B3:707 KIT:775 TGFBR2:827 ZFPM1:1092 KITLG:1312
Embryonic Organ Morphogenesis (GO:004856	0.15224434	42	6.433e−04	1.642e−01	SOX17:66 HOXC9:179 SIX1:295 HES1:544 NIPBL:561 OSR2:608
Ventricular Septum Development (GO:00032	0.16462001	36	6.330e−04	1.642e−01	TBX5:112 NKX2−5:323 HES1:544 TGFBR2:827 ZFPM1:1092 GATA4:1282
RNA Catabolic Process (GO:0006401)	0.12978303	57	7.059e−04	1.659e−01	SUPV3L1:140 LSM2:425 DXO:443 LSM5:528 RNASE6:594 SLFN14:715
Regulation Of Interleukin−2 Production (	0.14510264	45	7.616e−04	1.716e−01	RPS3:272 CD80:284 SPTBN1:360 CD86:415 FOXP3:551 PRKD2:700
Negative Regulation Of Lipid Catabolic P	−0.22684988	18	8.630e−04	1.866e−01	APOC1:549 AKT1:1013 PDE3B:1422 CIDEC:1566 ALK:2011 HCAR2:2901
Proton Motive Force−Driven Mitochondrial	0.13432855	50	1.022e−03	2.125e−01	NDUFS2:234 ATP5F1C:316 NDUFA12:475 NDUFB10:502 NDUFB8:797 NDUFB3:906
Detection Of Chemical Stimulus Involved	−0.08378033	127	1.131e−03	2.264e−01	OR2A14:8 OR10G9:16 OR2AG1:85 OR13C8:107 OR2T2:126 OR10G7:133
Androgen Biosynthetic Process (GO:000670	0.33001559	8	1.227e−03	2.324e−01	SRD5A2:84 HSD17B6:356 HSD17B3:1716 MED1:3247 HSD3B1:3470 HSD3B2:3744
Detection Of Chemical Stimulus Involved	−0.08242506	129	1.246e−03	2.324e−01	OR2A14:8 OR10G9:16 OR2AG1:85 OR13C8:107 OR2T2:126 OR10G7:133
Sensory Perception Of Taste (GO:0050909)	−0.23116760	16	1.368e−03	2.466e−01	CALHM3:21 TAS1R3:75 ASIC2:321 PKD2L1:703 GNAT3:1091 CALHM1:1450
Aortic Valve Development (GO:0003176)	0.14669998	39	1.529e−03	2.474e−01	TNFRSF1A:210 NKX2−5:323 SNAI1:451 TGFBR2:827 DLL4:830 BMP4:1269
Ketone Biosynthetic Process (GO:0042181)	0.30457354	9	1.556e−03	2.474e−01	SRD5A2:84 CYP11B1:128 CYP11B2:1001 SIRT6:2992 CACNA1H:3758 TPI1:4424
Negative Regulation Of Nucleic Acid−Temp	0.04434533	444	1.445e−03	2.474e−01	CHD3:28 ZNF24:31 LIMD1:48 NONO:62 USP47:83 YEATS2:198
Regulation Of Proteolysis Involved In Pr	0.19088224	23	1.532e−03	2.474e−01	GPX1:599 PSME3:1181 PINK1:1976 RHBDF1:2316 DNAAF4:2387 TMF1:2648
Negative Regulation Of Oxidoreductase Ac	−0.28796939	10	1.614e−03	2.494e−01	DRD5:523 HDAC6:592 PRKN:1827 ETFBKMT:2014 GZMA:2147 MT3:4373
Cardiac Ventricle Development (GO:000323	0.21338044	18	1.725e−03	2.521e−01	TBX5:112 HES1:544 DLL4:830 GATA4:1282 MDM4:1921 GJA5:1980
Negative Regulation Of Macromolecule Met	0.06756721	182	1.706e−03	2.521e−01	PARP9:8 ITGB8:47 POU4F1:150 FGF19:331 CNPY2:500 SFRP1:501
Regulation Of Activated T Cell Prolifera	0.15957636	32	1.788e−03	2.544e−01	TMIGD2:99 RPS3:272 FOXP3:551 CD274:663 RIPK3:1276 RC3H1:1430
His−Purkinje System Development (GO:0003	0.39912883	5	1.995e−03	2.587e−01	TBX5:112 NKX2−5:323 HOPX:1284 ID2:2688 IRX3:4919 NA
Homologous Chromosome Pairing At Meiosis	−0.16297479	30	2.009e−03	2.587e−01	KASH5:579 TEX11:619 RNF212:723 HORMAD1:794 SYCP2:1441 SYCP1:1763

EnrichmentHsSymbolsFile2 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_KERATINIZATION	−0.18361057	214	2.275e−20	1.478e−16	DSG4:1 KRT26:2 KRT2:3 KRT76:4 KRT34:9 KRT75:20
REACTOME_FORMATION_OF_THE_CORNIFIED_ENVE	−0.18302963	128	8.844e−13	2.872e−09	DSG4:1 KRT26:2 KRT2:3 KRT76:4 KRT34:9 KRT75:20
REACTOME_OLFACTORY_SIGNALING_PATHWAY	−0.09303375	390	3.081e−10	5.001e−07	OR2A14:8 OR10G9:16 OR51T1:51 OR51V1:62 OR8D4:66 OR56A5:70
REACTOME_SENSORY_PERCEPTION	−0.07559797	605	2.428e−10	5.001e−07	OR2A14:8 OR10G9:16 CALHM3:21 OR51T1:51 OR51V1:62 OR8D4:66
KEGG_OLFACTORY_TRANSDUCTION	−0.09227588	380	7.067e−10	9.179e−07	OR2A14:8 OR10G9:16 OR51T1:51 OR51V1:62 OR8D4:66 OR56A5:70
GRAESSMANN_APOPTOSIS_BY_DOXORUBICIN_DN	0.04085370	1705	2.721e−08	2.945e−05	ARMT1:19 COL1A2:26 CCN5:30 CEMIP2:50 GABRB2:51 NONO:62
RODRIGUES_THYROID_CARCINOMA_POORLY_DIFFE	0.06421325	608	7.012e−08	6.505e−05	CSNK1G3:16 PDCD5:42 USP47:83 BMS1:102 GLRX:105 RTCA:114
DODD_NASOPHARYNGEAL_CARCINOMA_UP	−0.04015077	1582	1.277e−07	1.037e−04	LGALS7:29 SERPINB11:32 KRT40:46 ABCA13:54 ALDH3B2:98 KCNJ15:103
BENPORATH_NANOG_TARGETS	0.04826707	951	5.238e−07	3.780e−04	HNRNPA1:78 ANKRD1:113 MTFR2:120 INO80C:145 KLF5:147 POU4F1:150
ACEVEDO_LIVER_TUMOR_VS_NORMAL_ADJACENT_T	0.05082543	826	7.733e−07	5.022e−04	GTF2IRD2B:11 GTF2IRD2:18 HSPH1:38 PDCD5:42 H2BC21:46 PIGB:65
MARSON_BOUND_BY_FOXP3_STIMULATED	0.04530700	974	1.907e−06	9.525e−04	HSPH1:38 ABHD8:56 KCTD10:71 FRMD8:77 MYNN:97 WRAP53:132
BUYTAERT_PHOTODYNAMIC_THERAPY_STRESS_UP	0.04979126	801	1.832e−06	9.525e−04	GOLGB1:15 ZNF24:31 HSPH1:38 GADD45A:39 CSNK2A1:79 AKAP8L:106
ACEVEDO_LIVER_CANCER_UP	0.04662380	921	1.809e−06	9.525e−04	FNDC3B:6 GTF2IRD2B:11 GTF2IRD2:18 COL1A2:26 H2BC21:46 GADD45GIP1:53
BENPORATH_SOX2_TARGETS	0.05226761	706	2.417e−06	1.121e−03	H2BC21:46 HNRNPA1:78 USP49:81 MYNN:97 ANKRD1:113 SGTA:123
RODRIGUES_THYROID_CARCINOMA_ANAPLASTIC_U	0.05298271	681	2.633e−06	1.140e−03	CSNK1G3:16 LACTB:82 USP47:83 GLRX:105 ANKRD1:113 RTCA:114
MARTENS_TRETINOIN_RESPONSE_DN	0.04914749	751	4.955e−06	2.011e−03	ZNF628:1 PHOX2A:22 H2BC21:46 NONO:62 TRMT10C:63 ABT1:73
MEISSNER_BRAIN_HCP_WITH_H3K27ME3	0.08078868	258	8.097e−06	3.093e−03	PHOX2A:22 HOXC13:80 IRX6:271 SIX1:295 NKX2−5:323 WDR86:334
FLECHNER_BIOPSY_KIDNEY_TRANSPLANT_REJECT	0.05460865	535	1.636e−05	5.901e−03	ALDH5A1:44 ABCC6:52 NONO:62 CSNK2A1:79 GLRX:105 ATP2A2:155
GABRIELY_MIR21_TARGETS	0.07456580	278	1.928e−05	6.588e−03	PARP9:8 ITGB8:47 CEMIP2:50 USP47:83 XKR3:138 KLF5:147
WONG_ADULT_TISSUE_STEM_MODULE	0.04715381	700	2.272e−05	7.377e−03	ABHD8:56 COL4A2:86 IL11RA:133 ANTXR2:152 ATP2A2:155 CEBPD:156
PID_MYC_REPRESS_PATHWAY	0.15761979	60	2.420e−05	7.482e−03	COL1A2:26 GADD45A:39 CEBPD:156 DNMT3A:300 SFRP1:501 TMEM126A:653
WP_GPCRS_CLASS_A_RHODOPSINLIKE	−0.07615207	253	3.098e−05	9.145e−03	MTNR1A:53 OR2AG1:85 ADORA2B:90 HTR4:91 OPN1MW:132 OPRL1:213
REACTOME_SIGNALING_BY_INTERLEUKINS	0.05617877	463	3.582e−05	9.305e−03	COL1A2:26 GRB2:94 IL11RA:133 CEBPD:156 PTPN12:158 NANOG:196
GEORGES_TARGETS_OF_MIR192_AND_MIR215	0.04216931	843	3.470e−05	9.305e−03	FNDC3B:6 ALDH5A1:44 CEMIP2:50 NOL12:92 PPM1A:115 PLOD1:185
MITSIADES_RESPONSE_TO_APLIDIN_UP	0.05930365	415	3.532e−05	9.305e−03	GADD45A:39 AKAP8L:106 TSPYL2:111 ASCC1:175 POLR2A:178 TTC17:227
NIKOLSKY_BREAST_CANCER_7P22_AMPLICON	−0.19206421	37	5.283e−05	1.183e−02	CYP2W1:39 GNA12:202 MICALL2:361 TMEM184A:859 FAM20C:921 SNX8:1318
LEE_LIVER_CANCER_MYC_E2F1_UP	0.15949480	54	5.038e−05	1.183e−02	COL1A2:26 IGFBP1:202 ANXA5:332 GPC3:630 PLSCR1:745 SLC1A4:864
HOLLERN_SQUAMOUS_BREAST_TUMOR	−0.09007686	170	5.144e−05	1.183e−02	KRT34:9 GJB4:17 KRT75:20 KRT83:24 KRT27:31 SERPINB11:32
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.04486473	713	4.774e−05	1.183e−02	KPNA7:25 COL1A2:26 GRB2:94 FLT3LG:117 IL11RA:133 CEBPD:156
GRAESSMANN_RESPONSE_TO_MC_AND_DOXORUBICI	0.04357760	740	5.803e−05	1.256e−02	CCN5:30 ZBTB12:68 CSNK2A1:79 U2AF1L4:91 MYNN:97 CEP41:135
LEE_LIVER_CANCER_E2F1_UP	0.14665686	61	7.464e−05	1.563e−02	IGFBP1:202 ANXA5:332 RHOB:519 JCAD:522 GPC3:630 COL5A2:742
BIOCARTA_SRCRPTP_PATHWAY	0.34314552	11	8.110e−05	1.646e−02	CDC25C:64 GRB2:94 CDC25A:623 CCNB1:1041 CSK:1139 CDC25B:1592
MOOTHA_MITOCHONDRIA	0.05490777	438	8.436e−05	1.660e−02	MTCH1:32 SLC25A37:87 CYP11B1:128 SUPV3L1:140 SLC30A9:171 FPGS:191
NABA_SECRETED_FACTORS	0.06190193	339	9.210e−05	1.759e−02	WNT1:2 CCL8:20 HCFC2:95 FLT3LG:117 LTB:165 S100A2:193
MARTENS_TRETINOIN_RESPONSE_UP	−0.04099227	790	9.535e−05	1.769e−02	KRT2:3 KRT34:9 GJB4:17 LMO1:23 KRT83:24 TREML4:35
REACTOME_MAPK_FAMILY_SIGNALING_CASCADES	0.06349183	319	9.899e−05	1.786e−02	GRB2:94 SHOC2:98 FLT3LG:117 SPTBN2:125 VCL:242 PSME4:259
BENPORATH_MYC_MAX_TARGETS	0.04178303	749	1.056e−04	1.853e−02	H2BC21:46 CLCN2:60 ABT1:73 HNRNPA1:78 HCFC2:95 GNL1:201
GUILLAUMOND_KLF10_TARGETS_UP	0.16059470	48	1.186e−04	2.019e−02	MT1X:423 STOM:456 SYT11:546 ALDH18A1:961 SAA2:1081 CCKBR:1520
REACTOME_AMINE_LIGAND_BINDING_RECEPTORS	−0.17348509	41	1.213e−04	2.019e−02	TAAR9:6 TAAR5:19 HTR4:91 HTR1D:249 GPR143:327 HTR1B:492
REACTOME_MITOCHONDRIAL_TRANSLATION	0.11512688	93	1.251e−04	2.031e−02	GADD45GIP1:53 GFM2:288 MRPL54:328 MRPL47:359 MRPL41:384 MRPL2:513

DisGeNET Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Diabetic Nephropathy	0.05651220	517	1.244e−05	4.585e−02	COL1A2:26 TSPYL2:111 ANKRD1:113 HNF4A:263 CD59:282 UTS2R:352
Generalized hypotonia	0.05141544	603	1.867e−05	4.585e−02	COL1A2:26 ALDH5A1:44 DHTKD1:61 NONO:62 TRMT10C:63 CSNK2A1:79
Malignant neoplasm of prostate	0.02721585	2864	6.445e−06	4.585e−02	WNT1:2 DIRAS3:13 CHD3:28 GADD45A:39 GADD45GIP1:53 CDC25C:64
Prostate carcinoma	0.02620214	2757	1.859e−05	4.585e−02	WNT1:2 DIRAS3:13 GADD45A:39 CEMIP2:50 GADD45GIP1:53 CDC25C:64
Acquired scoliosis	0.06044933	398	3.808e−05	7.480e−02	WNT1:2 COL1A2:26 ABCC6:52 NONO:62 TBX5:112 PLOD1:185
Joint laxity	0.12087824	90	7.502e−05	1.228e−01	COL1A2:26 NONO:62 PLOD1:185 RYR1:505 PAX2:555 HNRNPH2:720
Curvature of spine	0.05779455	389	9.844e−05	1.268e−01	WNT1:2 COL1A2:26 ABCC6:52 NONO:62 TBX5:112 PLOD1:185
Pulmonary Stenosis	0.14042393	64	1.033e−04	1.268e−01	SHOC2:98 G6PC3:143 CLIP2:277 GDF1:294 GTF2I:486 GPC3:630
Thin lips	0.13794006	65	1.213e−04	1.323e−01	GDF1:294 NKX2−5:323 HDAC8:518 NIPBL:561 NOTCH2:1077 ALG1:1093
Leukemia, T−Cell	0.06128492	324	1.583e−04	1.555e−01	HNRNPA1:78 DNMT3A:300 TYR:315 MAGI1:409 SNAP25:419 LYL1:538
Leukemia, Myelocytic, Acute	0.02858848	1511	2.725e−04	2.230e−01	WNT1:2 ABCC6:52 GADD45GIP1:53 CDC25C:64 SOX17:66 HOXC13:80
Mitochondrial Diseases	0.05599616	363	2.639e−04	2.230e−01	DHTKD1:61 TRMT10C:63 LACTB:82 COX4I2:89 PDSS1:144 PDSS2:154
Hyperreflexia	0.06111927	293	3.350e−04	2.301e−01	SPTBN2:125 NDUFS2:234 RUBCN:273 CLIP2:277 PEX16:280 PYCR2:339
Osteosarcoma	0.03581120	897	3.222e−04	2.301e−01	WNT1:2 DIRAS3:13 COL1A2:26 GADD45A:39 PDCD5:42 APOD:74
Renal Cell Dysplasia	0.29797520	12	3.515e−04	2.301e−01	PAX2:555 CITED1:619 ESR1:638 NOTCH2:1077 BMP4:1269 JAG1:1358
Down−sloping shoulders	0.20241100	24	5.991e−04	3.269e−01	CLIP2:277 GTF2I:486 MLXIPL:755 ELN:1288 PAX1:1361 LIMK1:1945
Embryonal Carcinoma	0.07627779	171	5.938e−04	3.269e−01	SOX17:66 CREG1:130 NANOG:196 VCL:242 MAP3K14:296 PODXL:362
Glioma	0.02456327	1903	5.489e−04	3.269e−01	WNT1:2 DIRAS3:13 CCL8:20 GADD45A:39 PDCD5:42 ITGB8:47
Congenital Abnormality	0.04198249	571	6.619e−04	3.287e−01	PHOX2A:22 PROP1:45 SOX17:66 TSPYL2:111 TBX5:112 ARFGEF2:121
Mammary Neoplasms	0.02287900	2173	6.694e−04	3.287e−01	WNT1:2 DIRAS3:13 COL1A2:26 HSPH1:38 GADD45A:39 LIMD1:48
Carcinogenesis	0.01842910	3580	9.445e−04	3.944e−01	WNT1:2 FNDC3B:6 COL1A2:26 CHD3:28 ZNF24:31 GADD45A:39
Elfin facies	0.30173030	10	9.531e−04	3.944e−01	CLIP2:277 GTF2I:486 ELN:1288 INSR:1824 LIMK1:1945 GTF2IRD1:2247
Fibrosarcoma	0.05768282	283	8.750e−04	3.944e−01	TBX5:112 CREG1:130 ASCC1:175 TNFRSF1A:210 SNAI1:451 FOXP3:551
Osteopenia	0.05420909	316	9.637e−04	3.944e−01	WNT1:2 COL1A2:26 PROP1:45 COX4I2:89 ANTXR2:152 LRP4:188
Phonophobia	0.28535035	11	1.049e−03	4.121e−01	CLIP2:277 GTF2I:486 MLXIPL:755 ELN:1288 EP300:1767 LIMK1:1945
COLORBLINDNESS, PARTIAL, DEUTAN SERIES	−0.16241756	32	1.478e−03	4.185e−01	OPN1MW:132 OPN1MW3:189 OPN1MW2:201 COX8A:809 HLA−DPB1:950 NOS1:1086
Body Height	0.05616316	281	1.242e−03	4.185e−01	FNDC3B:6 UGT1A9:40 ZNF341:127 POLR2A:178 MICA:250 DNMT3A:300
Bunion	0.16444733	31	1.534e−03	4.185e−01	NONO:62 CLIP2:277 GTF2I:486 COL5A2:742 MLXIPL:755 COL1A1:923
Ground−glass opacification on pulmonary	−0.25426871	13	1.502e−03	4.185e−01	SFTPA1:1281 SFTPA2:1454 SFTPC:1631 MUC5B:2714 FAM13A:3018 STN1:4164
Hallux Valgus	0.16893020	30	1.366e−03	4.185e−01	NONO:62 CLIP2:277 GTF2I:486 COL5A2:742 MLXIPL:755 COL1A1:923
Honeycomb lung	−0.25426871	13	1.502e−03	4.185e−01	SFTPA1:1281 SFTPA2:1454 SFTPC:1631 MUC5B:2714 FAM13A:3018 STN1:4164
Hyperactive behavior	0.03690239	665	1.287e−03	4.185e−01	ALDH5A1:44 GRB2:94 SHOC2:98 C1QL1:183 SNRPN:217 PRODH:219
Leukemogenesis	0.03688510	659	1.358e−03	4.185e−01	GADD45A:39 HNRNPA1:78 CXCR2:88 FLT3LG:117 DDR1:129 LTB:165
Liver carcinoma	0.01873934	3081	1.418e−03	4.185e−01	WNT1:2 FNDC3B:6 PARP9:8 ADH4:10 DIRAS3:13 COL1A2:26
Proteinuria	0.07190904	173	1.128e−03	4.185e−01	PDSS2:154 TREX1:215 PRODH:219 HNF4A:263 CLIP2:277 ACKR1:478
Reticular pattern on pulmonary HRCT	−0.25426871	13	1.502e−03	4.185e−01	SFTPA1:1281 SFTPA2:1454 SFTPC:1631 MUC5B:2714 FAM13A:3018 STN1:4164
Mental and motor retardation	0.03030300	971	1.590e−03	4.219e−01	WNT1:2 ALDH5A1:44 PROP1:45 DHTKD1:61 CSNK2A1:79 COL4A2:86
Bicuspid aortic valve	0.12582501	49	2.321e−03	4.286e−01	PLOD1:185 NKX2−5:323 MLXIPL:755 TGFBR2:827 ACTA2:1209 ELN:1288
Bowel diverticula	0.35882531	6	2.335e−03	4.286e−01	COL5A2:742 COL1A1:923 ALDH18A1:961 ELN:1288 FBLN5:1972 COL5A1:9821
Bowel diverticulosis	0.35882531	6	2.335e−03	4.286e−01	COL5A2:742 COL1A1:923 ALDH18A1:961 ELN:1288 FBLN5:1972 COL5A1:9821

MGI_Mammalian_Phenotype_Level_4 Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
MP0002086 abnormal extraembryonic tissu	0.06435275	490	2.442e−06	1.162e−03	ITGB8:47 GADD45GIP1:53 SOX17:66 COL4A2:86 GRB2:94 KLF5:147
MP0002085 abnormal embryonic tissue	0.05206809	661	1.337e−05	1.817e−03	WNT1:2 ITGB8:47 SOX17:66 CSNK2A1:79 GRB2:94 TBX5:112
MP0002082 postnatal lethality	0.04388809	931	2.290e−05	1.817e−03	PHOX2A:22 COL1A2:26 ALDH5A1:44 PROP1:45 ITGB8:47 GADD45GIP1:53
MP0010770 preweaning lethality	0.04388809	931	2.290e−05	1.817e−03	PHOX2A:22 COL1A2:26 ALDH5A1:44 PROP1:45 ITGB8:47 GADD45GIP1:53
MP0010769 abnormal survival	0.04321357	1021	1.533e−05	1.817e−03	PHOX2A:22 COL1A2:26 ALDH5A1:44 PROP1:45 ITGB8:47 GADD45GIP1:53
MP0010768 mortality/aging	0.04292410	1044	1.477e−05	1.817e−03	PHOX2A:22 COL1A2:26 ALDH5A1:44 PROP1:45 ITGB8:47 GADD45GIP1:53
MP0000001 mammalian phenotype	0.03216815	3058	3.377e−05	2.009e−03	FNDC3B:6 PHOX2A:22 COL1A2:26 GADD45A:39 ALDH5A1:44 PROP1:45
MP0001614 abnormal blood vessel	0.04930862	673	3.246e−05	2.009e−03	ITGB8:47 ABCC6:52 COL4A2:86 CXCR2:88 SHOC2:98 DDR1:129
MP0000266 abnormal heart morphology	0.04670393	715	5.355e−05	2.832e−03	ITGB8:47 CSNK2A1:79 COL4A2:86 GRB2:94 SHOC2:98 TBX5:112
MP0003984 embryonic growth retardation	0.06052890	338	1.868e−04	8.894e−03	GADD45GIP1:53 SOX17:66 COL4A2:86 SHOC2:98 PTPN12:158 P4HA1:214
MP0002135 abnormal kidney morphology	0.05167133	457	2.460e−04	1.022e−02	GADD45A:39 ABCC6:52 CXCR2:88 CYP11B1:128 PDSS2:154 LRP4:188
MP0002088 abnormal embryonic growth/wei	0.05864093	345	2.577e−04	1.022e−02	GADD45GIP1:53 SOX17:66 COL4A2:86 SHOC2:98 PTPN12:158 P4HA1:214
MP0002925 abnormal cardiovascular devel	0.04748813	521	3.499e−04	1.281e−02	ITGB8:47 CSNK2A1:79 CXCR2:88 GRB2:94 SHOC2:98 TBX5:112
MP0003632 abnormal nervous system	0.10847901	91	3.778e−04	1.285e−02	MAP3K12:167 TNFRSF1A:210 CD86:415 LEPR:849 ITGB1:883 LAMA5:1400
MP0000598 abnormal liver morphology	0.04633164	483	7.486e−04	2.238e−02	SOX17:66 CXCR2:88 CYP11B1:128 PTPN12:158 LTB:165 XIRP2:182
MP0002080 prenatal lethality	0.03022975	1353	7.991e−04	2.238e−02	WNT1:2 ITGB8:47 GADD45GIP1:53 SOX17:66 CSNK2A1:79 COL4A2:86
MP0002114 abnormal axial skeleton	0.04859470	433	7.678e−04	2.238e−02	COL1A2:26 HOXC13:80 TBX5:112 SUPV3L1:140 HOXC9:179 SIX1:295
MP0005380 embryogenesis phenotype	0.04833790	419	9.797e−04	2.454e−02	GADD45GIP1:53 SOX17:66 COL4A2:86 SHOC2:98 SUPV3L1:140 PTPN12:158
MP0001672 abnormal embryogenesis/ devel	0.04833790	419	9.797e−04	2.454e−02	GADD45GIP1:53 SOX17:66 COL4A2:86 SHOC2:98 SUPV3L1:140 PTPN12:158
MP0010630 abnormal cardiac muscle	0.06481294	217	1.186e−03	2.823e−02	CSNK2A1:79 HIF3A:141 KLF5:147 ATP2A2:155 XIRP2:182 TREX1:215
MP0002128 abnormal blood circulation	0.04735776	410	1.387e−03	3.144e−02	COL1A2:26 ITGB8:47 CSNK2A1:79 COL4A2:86 GRB2:94 SHOC2:98
MP0003121 genomic imprinting	0.16291243	29	2.438e−03	5.275e−02	SNRPN:217 HDAC8:518 CITED1:619 GPC3:630 RB1:1410 MAGEL2:1446
MP0003861 abnormal nervous system	0.03472055	723	2.550e−03	5.277e−02	WNT1:2 GADD45A:39 ITGB8:47 CSNK2A1:79 CXCR2:88 PTPN12:158
MP0003935 abnormal craniofacial develop	0.05470741	247	3.596e−03	7.132e−02	ITGB8:47 CSNK2A1:79 GRB2:94 SUPV3L1:140 LRP4:188 VCL:242
MP0004185 abnormal adipocyte glucose	0.20943612	16	3.760e−03	7.159e−02	BGLAP:801 FABP5:1342 MARK2:1683 E2F1:1747 TNKS2:1753 CEBPA:1963
MP0000163 abnormal cartilage morphology	0.05295416	251	4.512e−03	8.260e−02	WNT1:2 LRP4:188 MATN1:223 THRA:249 SIX1:295 NKX3−2:341
MP0002113 abnormal skeleton development	0.04252240	387	5.184e−03	9.139e−02	FNDC3B:6 MATN1:223 THRA:249 SIX1:295 DMRT2:327 NKX3−2:341
MP0002127 abnormal cardiovascular syste	0.11195731	51	5.838e−03	9.924e−02	S1PR1:967 HBEGF:998 ZIC3:1022 GATA4:1282 CKMT2:1362 SOS1:1390
MP0002557 abnormal social/conspecific i	−0.05414955	213	7.259e−03	1.191e−01	HTT:14 ING1:95 CPLX1:143 NLGN2:204 AR:311 KCNA1:376
MP0000313 abnormal cell death	0.03661473	466	8.763e−03	1.390e−01	GADD45GIP1:53 USP47:83 GLRX:105 SESN2:146 ANTXR2:152 IGFBP1:202
MP0001697 abnormal embryo size	0.03828219	412	9.586e−03	1.472e−01	ITGB8:47 GADD45GIP1:53 P4HA1:214 VCL:242 HNF4A:263 SIX1:295
MP0004133 heterotaxia	0.13170888	32	1.007e−02	1.498e−01	GDF1:294 FOXH1:347 KIF3A:647 HHIP:844 ZIC3:1022 LAMA5:1400
MP0003122 maternal imprinting	0.18072627	16	1.241e−02	1.789e−01	SNRPN:217 GPC3:630 MAGEL2:1446 MEST:2099 PEG10:3426 IGF1R:3822
MP0001270 distended abdomen	0.09401555	58	1.362e−02	1.852e−01	SIX1:295 FOXP3:551 GPC3:630 KIT:775 RFX6:1114 CYS1:1154
MP0002928 abnormal bile duct	0.12656897	32	1.340e−02	1.852e−01	SOX17:66 MAP3K14:296 HES1:544 PTEN:813 HNF1B:999 NOTCH2:1077
MP0002138 abnormal hepatobiliary system	0.24520695	8	1.637e−02	2.012e−01	MAP3K14:296 HNF1B:999 PKHD1:1598 ABCB4:4074 SOCS2:4892 ACVRL1:5907
MP0003123 paternal imprinting	0.21912529	10	1.648e−02	2.012e−01	CITED1:619 GPC3:630 UBE3A:2747 HELLS:2994 CDKN1C:3696 RASGRF1:5344
MP0000767 abnormal smooth muscle	0.07657145	83	1.647e−02	2.012e−01	PLOD1:185 DLL4:830 SMAD5:955 S1PR1:967 GPR4:1034 NOTCH2:1077
MP0005620 abnormal muscle contractility	0.04908653	208	1.612e−02	2.012e−01	ATP2A2:155 VCL:242 THRA:249 NKX2−5:323 UTS2R:352 MYOZ2:385
MP0005367 renal/urinary system phenotyp	0.15637343	19	1.843e−02	2.088e−01	DMTF1:618 PTEN:813 BMP4:1269 PAX1:1361 SCARB2:1453 FOXC1:2254

tissue_specific Top pathways by non−permulation

Geneset	stat	num.genes	pval	p.adj	gene.vals
testis	−0.05900331	1688	7.504e−14	4.052e−12	KRT26:2 TEX53:5 FAM170A:10 CALHM3:21 CCNY:28 SIGLECL1:38
aorta	0.08065092	156	5.487e−04	8.121e−03	GADD45A:39 COL4A2:86 PDGFD:290 ITGA11:319 MYOZ2:385 PLD5:591
leg.skin	−0.05942235	288	6.015e−04	8.121e−03	KRT2:3 ALOXE3:13 GJB4:17 KRT75:20 LGALS7B:26 LGALS7:29
transformed.skin.fibroblast	0.06217847	276	4.362e−04	8.121e−03	COL1A2:26 HOXC9:179 PLOD1:185 DNAJB4:204 NEDD4:244 ANXA5:332
suprapubic.skin	−0.06087223	259	8.404e−04	9.077e−03	KRT2:3 ALOXE3:13 GJB4:17 KRT75:20 LMO1:23 LGALS7B:26
skeletal.muscle	0.05796059	258	1.506e−03	1.356e−02	MYLPF:54 CHRNA10:55 PLCD4:57 GLRX:105 ATP2A2:155 XIRP2:182
EBV.lymphocyte	0.03304267	757	2.704e−03	2.086e−02	CDC25C:64 MTFR2:120 ANKRD37:157 RHEBL1:163 P4HA1:214 ADPRH:241
tibial.artery	0.07899640	114	3.718e−03	2.510e−02	CCN5:30 WTIP:43 LPP:371 RHOB:519 TPM2:734 KBTBD13:882
bladder	−0.13374723	30	1.131e−02	5.554e−02	DHRS2:55 UPK1B:295 SIGLEC15:398 ACSM6:475 PLA2G2F:501 CYP4B1:1236
spleen	−0.03943545	367	1.055e−02	5.554e−02	TAS1R3:75 CEACAM21:174 ARHGAP25:466 GRAPL:488 FCRL2:506 P2RY13:512
eye.development	0.09048437	69	9.528e−03	5.554e−02	TBX5:112 KLF5:147 TYR:315 SFRP1:501 HES1:544 PAX2:555
vagina	−0.06450951	119	1.553e−02	6.989e−02	LGALS7B:26 LGALS7:29 SERPINB11:32 GJB5:58 TMPRSS11F:60 ALDH3B2:98
left.ventricle	0.08308851	69	1.728e−02	7.180e−02	MYZAP:100 ANKRD1:113 NKX2−5:323 COX6A2:373 MYOZ2:385 ASB15:807
esophagus.mucosa	−0.04108952	263	2.316e−02	8.770e−02	LGALS7B:26 LGALS7:29 SERPINB11:32 GJB5:58 TMPRSS11F:60 ALDH3B2:98
liver	0.03336432	399	2.436e−02	8.770e−02	ADH4:10 UGT1A9:40 ABCC6:52 TM6SF2:93 TFR2:136 HSD11B1:139
cerebellum	0.03072503	409	3.593e−02	1.212e−01	TMEM59L:118 SPTBN2:125 MAP3K12:167 LZTS3:216 HCN3:306 RTL5:337
minor.salivary.gland	−0.05460680	91	7.276e−02	2.311e−01	MUC16:307 ODAM:433 CXCL5:892 KRT6B:979 RAET1L:1009 KRT81:1113
esophagogastric.junction	0.16531780	9	8.598e−02	2.579e−01	BARX1:1020 GADL1:3465 COL4A5:4077 HOXA4:4416 F2RL2:6747 ADCY5:8813
peyers.patch	−0.03294281	199	1.118e−01	3.179e−01	KRT20:48 HTR4:91 NR1I2:108 MUC13:276 SLC17A8:289 BCO1:332
pituitary.gland	0.02711169	234	1.568e−01	4.234e−01	CLUL1:148 SPRED3:261 SSTR5:275 SIX1:295 VEPH1:387 TUSC3:393
hypothalamus	−0.04552503	78	1.657e−01	4.262e−01	PNMA6F:65 RTP1:187 SYTL5:541 GNG8:720 PNMA5:743 LRRC55:761
adrenal.gland	0.03655117	116	1.758e−01	4.314e−01	CYP11B1:128 KLF14:168 TMEM150C:213 ALAD:324 GALM:400 GSTA2:586
cortex.kidney	0.03347312	132	1.864e−01	4.376e−01	UGT1A9:40 HNF4A:263 DMRT2:327 GDF15:552 PAX2:555 GSTA2:586
transverse.colon	−0.02809902	170	2.092e−01	4.707e−01	KRT20:48 NR1I2:108 MUC13:276 OMP:388 TBX10:442 TMEM238L:560
sigmoid.colon	0.04701163	53	2.373e−01	5.127e−01	PHOX2A:22 NKX3−2:341 PNCK:662 GUCY1A1:812 TES:907 HAND2:1012
nucleus.accumbens	−0.03177730	104	2.647e−01	5.498e−01	HTR1D:249 TTYH1:538 CNMD:837 LAMP5:967 HTR6:974 LYPD1:1378
Brodmann.area.24	−0.04920749	33	3.285e−01	6.486e−01	DOK6:214 DRD5:523 PTER:767 OSTN:1218 CSPG5:3055 STUM:3785
caudate.nucleus	−0.05350730	27	3.363e−01	6.486e−01	TAC1:1234 PSD2:1297 MLC1:1560 MAPK4:1938 HS3ST5:2004 FAM237A:2074
Brodmann.area.9	−0.01625361	187	4.466e−01	7.340e−01	CPLX1:143 RTP1:187 SNTG1:247 MCHR2:280 PNMA6A:460 MAS1:514
ectocervix	0.04321139	27	4.375e−01	7.340e−01	ESR1:638 CPXM1:843 ADAMTSL5:969 LRFN5:1107 FGF7:1706 MATN2:1801
greater.omentum	0.03706103	38	4.298e−01	7.340e−01	SOX17:66 HBEGF:998 BARX1:1020 GPR4:1034 GFPT2:1503 SPRY1:1811
pancreas	−0.02216387	118	4.077e−01	7.340e−01	IAPP:25 REG3G:180 ACSM6:475 TEX11:619 CPA2:722 SLC30A2:990
stomach	−0.02212265	99	4.486e−01	7.340e−01	VWA2:33 CAPN8:151 CYSTM1:430 PSAPL1:474 SLC9A4:531 CXCL17:697
prostate	0.02801576	53	4.813e−01	7.426e−01	SRD5A2:84 H2BC5:262 P2RX2:313 TULP1:970 KLK2:1407 EPHA3:1528
thyroid	0.01587078	177	4.694e−01	7.426e−01	CLUL1:148 WDR86:334 BSPRY:693 GAL3ST3:865 TLE6:931 TSTD1:987
hippocampus.proper	0.04783273	13	5.506e−01	8.258e−01	CNIH2:901 NTS:3133 SLC17A7:4051 OLIG2:4245 CTXND1:5669 NEUROG3:5716
substantia.nigra	−0.04799683	11	5.816e−01	8.488e−01	SLC6A11:746 SPX:1054 TH:1686 RET:3960 FOXB1:6322 CHRM5:9825
atrium.auricle	0.01500914	96	6.126e−01	8.705e−01	TBX5:112 ANKRD1:113 SHD:173 DKK3:218 NKX2−5:323 COX6A2:373
esophagus.muscularis.mucosa	0.02643015	28	6.287e−01	8.705e−01	S1PR3:537 JPH2:1334 PALLD:3215 NTN1:3445 GADL1:3465 RGS2:3597
uterus	0.01607189	63	6.598e−01	8.908e−01	KLHL13:24 ESR1:638 IL17RD:661 RSPO1:704 LIPI:964 CHRDL2:1226